

SHORT COMMUNICATION

Antifungal activity of tea tree oil from *Melaleuca alternifolia* against *Trichophyton equinum*: An *in vivo* assayF. Pisseri^a, A. Bertoli^b, S. Nardoni^c, L. Pinto^c, L. Pistelli^b, G. Guidi^d, F. Mancianti^{c,*}^a*Scuola di Omeopatia Cimi-koinè, Roma, Italy*^b*Dipartimento di Chimica Bioorganica e Biofarmacia, Università di Pisa, Italy*^c*Dipartimento di Patologia Animale, Igiene e Profilassi degli Alimenti, Università di Pisa, Italy*^d*Dipartimento di Clinica Veterinaria, Università di Pisa, Italy***Abstract**

Dermatophytes are a group of keratinophilic and keratinolytic molds, some of which are responsible for ringworm. Among them *Trichophyton equinum*, which mostly infects equids, can cause extensive outbreaks in stud farms. The conventional treatment of equine trichophytosis is topical, based upon medicated shampoos to reduce the spread of infection among the animals. Nevertheless the popularity of phytotherapy is at an all-time peak, and the interest for natural alternatives or complements to conventional drug therapy is challenging both in human and veterinary field. Among herbal remedies Tea Tree Oil (TTO) shows a wide range of antimicrobial activities. A randomized open clinical trial was carried out on 60 thoroughbred breeding horses affected by equine ringworm. The animals were randomly divided into 2 groups of 30 subjects. Diagnostic criteria were the presence of clinical signs and positive *T. equinum* culture. Specificity control using TTO mixture in 5 not dermatophyte affected animals was achieved also. The antimycotic activity against *T. equinum* of a mixture containing 25% TTO in sweet almond oil, was evaluated *in vivo* treating 30 subjects, the others were administered enilconazole 2% solution. The animals of both groups were topically treated twice a day for 15 days with a 25% mixture of TTO diluted in sweet almond oil and every 3 days, four times with enilconazole rinses, respectively. The clinical and mycological outcome were evaluated at day 30 from the start of the treatments. Data analysis was performed by chi square test. All the treated animals showed complete clinical and aetiological healing. Part of control subjects also, showed an improvement and none of them exacerbated the lesions. This therapeutic protocol appears to be effective and versatile, being applicable immediately after physical examination, prior to have the laboratory response. It could be an alternative for practitioners interested in herbal medicines, contributing to fulfill the gap existing between *in vitro* and clinical studies.

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Keywords: *Trichophyton equinum*; Tea tree oil (TTO); *Melaleuca alternifolia*; Antifungal activity; *In vivo***Introduction**

Dermatophytes are a group of morphologically and physiologically related molds some of which cause well-

defined infections: dermatophytoses (tineas or ringworm). They are keratinophilic and keratinolytic. Based on their ability to digest keratin *in vitro* and/or *in vivo* these fungi can cause infectious or not-infectious disease. Zoophilic dermatophytes spend their life on animal keratin and don't live on hair-baits in the soil (Simpanya 2000), such as *Trichophyton equinum*, which

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has a narrow range of receptive hosts mostly infecting equids. This fungal species represents a threat to animal health. Infected horses present a clinical symptomatology more or less severe, depending on their innate and/or acquired defenses. During this period these subjects spread the dermatophyte in the environment and to receptive hosts, furthermore the fungal propagula can maintain their infectivity for several months, making the environment, the girths, all the tack and grooming equipment responsible for new infections. Ringworm is most common in young animals (aged under 3 years) or in old and debilitate subjects. The disease appears as small and circular patches of hair loss, with skin more or less inflamed and scabbed. They can spontaneously recover but an aetiological clearance may require several months. If untreated, the lesions spread leading to large and irregular areas with broken hairs, with blisters and crusts. The lesions are normally located on the face and around the eyes as well as around the girth and in saddle area. They are frequently painful making difficult to saddle the horse. Pruritus is a not constant feature.

In horses conventional treatment is topic, based upon medicated shampoos. Enilconazole is considered the first choice drug in conventional medicine, due to its efficacy, safety, and cheapness. Nevertheless the popularity of phytotherapy is at an all-time peak, and the interest for natural alternatives or complements to conventional drug therapy is challenging both in human and veterinary field. The return of interest in folk remedies and herbal medicines leads to an increased request for traditional belief systems with a consequent widespread usage for healing of domestic animals, also. For these reasons herbal remedia are more and more used in the treatment of skin diseases in animals and, among them, essential oils (EOs) appear to be particularly of interest.

The EO of *Melaleuca alternifolia* or Tea Tree Oil (TTO) has been used in Australian traditional medicine, and more recently worldwide, for its wide range of antimicrobial activity. Its chemical composition is strictly related to the quality of the raw plant material, therefore significant differences in the yields and composition of the TTO essential oil can be found. TTO has many therapeutical properties, such as anti-inflammatory and antimicrobial. Its pharmaceutical and medicinal aspects have been carefully reviewed by Saller et al. (1998) who reported the lack of clinical studies in this field. Due to its lipophylic nature which facilitates skin penetration TTO is used as local formulation for different dermatological disorders. The antimycotic effect of TTO in veterinary medicine has successfully been demonstrated against many strains of *Malassezia pachydermatis* from seborrhoeic dermatitis (Weseler et al. 2002). Several TTO-based topical products are purchased in the traditional use, but to the best of our knowledge, it has never been essayed to evaluate the course of *T. equinum* natural infection in horses.

The aim of the present paper was to evaluate, by an open randomized trial, the antimycotic activity of TTO treating equine ringworm in naturally infected horses.

Materials and methods

The oils were provided by FLORA s.r.l., Lorenzana (PI), Italy.

A randomized open clinical trial to compare therapeutic effects of TTO with enilconazole in equine ringworm was carried out. Sixty thoroughbred breeding horses aged from 18 to 24 months with dermatological signs suggestive of equine ringworm were included. They were randomly divided into 2 groups of 30 subjects and were topically treated twice a day for 15 days with a 25% mixture of TTO (serial no. 70438) diluted in sweet almond oil (serial no. 70312). All the essential oil samples were analyzed by Gas-Chromatography Mass Spectrometry (GC-MS) before being used for the assays. The other animals were conventionally treated with enilconazole (Imaverol[®], Janssen) 2% solution, washing the affected and surrounding areas every 3 days, four times. Animals with clinical signs not referable to ringworm and/or medically treated during the last 2 months were not included into the study. Furthermore only subjects whose lesions were lasting from 10 to 30 days were admitted.

Mycotic culture, microscopic examination and clinical signs were considered diagnostic criteria.

More frequent clinical features consisted of crusting and scaling alopecic areas, with patchy hair loss without significant itchiness. The lesions were mostly located on the back, less frequently on the face and on the legs. All the symptomatic animals had recently been introduced in the stud farm from the UK and from the Ireland.

Five animals affected by horse summer eczema were used as control group. Considering the high infectivity of *T. equinum* untreated animals were not included.

TTO mixture was used to soak the crusts, while enilconazole was administered after crusts removal. To reduce the risk of re-infection and contamination by ringworm spores, all stable equipment were thoroughly washed and bedding was destroyed.

Mycological and clinical follow-up were performed on day 30 from the start of the treatment. The samples were drawn by McKenzie's technique, and seeded onto selective medium for dermatophytes added with nicotinic acid 10%. The plates were incubated at 25 °C for one month. A clinical score (excellent = complete resolution, good = partial resolution, inadequate = lack of recovery) was assigned to each foal to evaluate the clinical outcome.

Data were compared by a chi square test.

Results and discussion

All treated animals showed a good ($n = 6$) to excellent ($n = 24$) clinical score, with complete healing of cutaneous lesions and partial (good) or total (excellent) regrowth of hair. The control group treated with enilconazole showed an excellent (21 subjects) and a good outcome (9 subjects). Three out of 5 animals of the control group had an improvement of their clinical condition, while the others did not exacerbate the lesions. All the cultures as well as hair microscopic examinations carried out after the treatment, scored negative for dermatophytes.

No significant differences between the 2 groups regarding the effectiveness of the treatments were noted.

There are many pitfalls in using herbs, the method of preparation and time of harvest are very important, and the use of chromatography and other technologies to identify active compound peaks can help to choose the right plant species (Dattner 2003). Equine ringworm is a self-healing infection. Nevertheless, a treatment is strongly requested to obtain a quicker healing of crusty and inflammatory lesions disturbing the horse tame and to limit the infection spreading among the animals in the same stable.

Enilconazole is the sole effective antimycotic drug against *T. equinum*, other conventional antimycotic drugs have a narrow spectrum of effect and can cause severe side effects, furthermore these treatments can induce pharmacoresistance and their administration must be avoided in large animals in zootechnical production. These molecules frequently are ineffective, as fungi can still be isolated from lesions for up to 15–30 days after treatment (Pascoe 1984) failing to reduce the spread of infection among the animals. However equine practitioners point to a more natural approach, recommending natural products instead of azoles for the ringworm's treatment. In this case TTO was effective against *T. equinum* in the *in vivo* assay. Any horse from control group did not worsen, and 3/5 subjects with summer eczema showed an improvement of clinical signs, while all the enilconazole treated animals were cured. Sweet almond oil was employed as an excipient in the remedium. It is considered a particularly suitable carrier for dermatological preparations applied to the skin or in the external ear, since it is well tolerated, easily absorbed and spreads to the dermic layer allowing an optimal absorption of the active principle. This oil is obtained through the cold pressing of *Prunus dulcis*'s seeds and it is traditionally well known as eudermic, with a marked moisturizing, emollient, lenitive anti-inflammatory and anti-itching activity (Pisseri et al. 2008). Some polyphenols with an antioxidant activity have been recently identified (Amico et al. 2006) in its composition and its polysaccharidic fraction has shown stimulating activity of lymphocytes of rat both *in vivo* and *in vitro*.

These reasons make it not a simple carrier only, since it actively contributes to improve the skin balance altered in dermatological diseases. Associated with TTO it can enhance a faster healing of fungal lesions.

TTO is also easy to use, not recommending the removal of crusts with a hard brush before treating, as requested with enilconazole washes. For these reasons, this therapeutic protocol appears to be versatile, being applicable immediately after physical examination, prior to have the laboratory response.

The use of standardized test conditions in this field is very complex to obtain. Furthermore, *in vivo*, the synergistic activity of multiple active principles present in TTO, unpredictable when a specific isolated component is studied, should also be considered.

In conclusion, essential oils from many plants are known to possess antifungal activity, but only limited information exists about activity towards human and animal fungal pathogens. EOs can be considered potential natural fungicides, alternative to synthetic products. In the present study a fungicidal concentration without adverse reactions was assessed on horse skin, following Saller et al. (1998) who recommend the need for scientific clinical studies in evaluating TTO topical application.

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